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积分

一. 多元积分的相关概念

积分 < 欧氏空间积分 (多重积分) ⟷ 将Riemann积分推广到 $\mathbb{R}^n$ 中
推广至一般的M上 (曲线.曲面积分)

1. 可积性理论

一元 f 可积 $\Longleftrightarrow$ f 有界且 $disc(f)$ 零测

多元 f 可积 $\Longleftarrow$ f 有界且 $disc(f)$. $disc(X_D)$ 皆零测 $\left(X_D(x)=\begin{cases}1 & x\in D\\ 0 & x\notin D\end{cases},\ 定义 f=f\cdot X_D\right)$

2. 定义: $D\subseteq\mathbb{R}^n$ 的 n 维 volume $vol(D)=\int\cdots\int_D 1$

3. 性质: ① $\int_D(\alpha f+\beta g)=\alpha\int_D f+\beta\int_D g$

② $\int_{D_1\cup D_2} f=\int_{D_1} f+\int_{D_2} f-\int_{D_1\cap D_2} f$

esp. if $D_1\cap D_2$ 是光滑的 n-1 维超曲面 (D_1, D_2 的公共边界), 则

$0=-M\,vol(D_1\cap D_2)\le\int_{D_1\cap D_2} f\le M\int_{D_1\cap D_2}1=M\,vol(D_1\cap D_2)=0$ (设 $|f|\le M$)

$\Rightarrow \int_{D_1\cup D_2} f=\int_{D_1} f+\int_{D_2} f$

③ 积分不等式 i) 设 $f(x)\le g(x)$, $\forall x\in D$, 则 $\int_D f\le\int_D g$

ii) $\left|\int_D f\right|\le\int_D|f|$

④ 积分中值Thm)

设 $D\subseteq\mathbb{R}^n$ 紧且道连, $disc(X_D)$ 零测, 则 $\forall f\in C(D)$, $\exists X_0\in D$, s.t.

$$\frac{\int_D f}{vol(D)}=f(X_0)$$

补注: 定义 f 在 I 上的 n 重积分: 记为 $\int\cdots\int_I f(x_1,\cdots,x_n)\,d\,vol=\int\cdots\int_I f(x_1,\cdots,x_n)\,dx_1\cdots dx_n$

$=\int\cdots\int_I f(x_1,\cdots,x_n)\,|dx_1\cdots dx_n|$ → 强调 vol 为正

$=\int_I f$ (越简略反而不易有歧义)

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一元 多元

N-L 分部 换元 ⟹ Stokes 换元 Fubini

二. 多重积分计算

1. Fubini: 设 $D \subseteq \mathbb{R}^2$ 有界, if 下式两边有定义, 则等式成立

$$\iint_D f = \int_a^b dx \left(\int_{\{x\}\times\mathbb{R}\cap D} f(x,y)\,dy \right) \qquad D \subseteq [a,b]\times[c,d]$$

(y 从小往大积)

推论: (在重积分习的条件下, 累次积分可自由换序)

设 f 在 D 上可积, 则

$$\int dx \left(\int_{\{x\}\times\mathbb{R}\cap D} f(x,y)\,dy \right) = \int dy \left(\int_{\{y\}\times\mathbb{R}\cap D} f(x,y)\,dx \right)$$

(y 从小往大积) (x 从小往大积)

n 重 Fubini: 设 $\Omega \subseteq \mathbb{R}^n$ 有界, f 在 Ω 上可积, 则

$$\int\cdots\int_\Omega f = \int\cdots\int dx_1\cdots dx_{n-1} \left(\int_{\{x_1,\cdots,x_{n-1}\}\times\mathbb{R}\cap D} f(x_1,\cdots,x_n)\,dx_n \right)$$

(x_n 从小往大积)

$$= \int dx_n \left(\int\cdots\int_{\{x_n\}\times\mathbb{R}^{n-1}\cap\Omega} f(x_1,\cdots,x_n)\,dx_1\cdots dx_{n-1} \right)$$

e.g.1 (费曼参数化) $V = \{(x_1,\cdots,x_n) \mid \begin{matrix} x_i \geq 0 \ \forall i \\ \sum x_i \leq 1 \end{matrix}\}$ 求 $I = \int\cdots\int_V \left(\dfrac{1}{\sum_{i=1}^n a_i x_i + a_0\left(1-\sum_{i=1}^n x_i\right)} \right)^{n+1}$

方法: 多重积分 → 累次积分时, 只需

对 V 的定义方程(不等式)时刻做同解变形, 即把 $x_1,\cdots,x_n$ 的约束依次同解变形

记 $F_n(a_0;\cdots,a_n) = \int\cdots\int_V \left(\dfrac{1}{\cdots}\right)^{n+1} dx_1\cdots dx_n$

Fubini $\int\cdots\int_{\substack{x_1+\cdots+x_{n-1}\leq 1 \\ x_i\geq 0}} dx_1\cdots dx_{n-1} \left(\int_0^{1-x_1-\cdots-x_{n-1}} \left[(a_1-a_0)x_1+\cdots+(a_n-a_0)x_n+a_0\right]^{-n-1} dx_n \right)$

$$= \frac{1}{n(a_n-a_0)}\left[F_{n-1}(a_0;a_1,\cdots,a_{n-1}) - F_{n-1}(a_n;a_1,\cdots,a_{n-1})\right]$$

$$= \frac{1}{n(a_0-a_n)}\left[\left(\sum_{i=1}^{n-1}(a_i-a_n)x_i+a_n\right)^{-n} - \left(\sum_{i=1}^{n-1}(a_i-a_0)x_i+a_0\right)^{-n}\right]\Bigg|_{\substack{x_n = 1-x_1-\cdots-x_{n-1} \\ x_n = 0}}$$

归纳来证 (猜:) $F_n(a_0;a_1\cdots,a_n) = \dfrac{1}{n!\,a_0\cdots a_n}$

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e.g.2 $\Omega = \begin{Bmatrix} x^2+y^2 \le 1 \\ x^2+z^2 \le 1 \\ y^2+z^2 \le 1 \end{Bmatrix}$ 求 $vol(\Omega) = \iiint_\Omega dxdydz$

$$\iiint_\Omega dxdydz = \iint_{x^2+y^2\le 1} dxdy \int_{\substack{z^2 \le 1-x^2 \text{且} \\ z^2 \le 1-y^2}} dz = \iint_{x^2+y^2\le 1} dxdy \int_{-\sqrt{\min}}^{+\sqrt{\min}} dz$$

$$= \iint_{x^2+y^2\le 1} 2\sqrt{\min\{1-x^2, 1-y^2\}}\, dxdy = 16 \iint_{\substack{0\le y\le x \\ x^2+y^2\le 1}} \sqrt{1-x^2}\, dxdy$$

$$= 16\int_0^{\frac{1}{\sqrt{2}}} \sqrt{1-x^2}\, dx \int_0^x dy = 16\int_0^{\frac{1}{\sqrt{2}}} x\sqrt{1-x^2}\, dx = 16 \left. (1-x^2)^{\frac{3}{2}} \Big/ \tfrac{3}{2}\cdot(-2) \right|_{x=0}^{x=\frac{1}{\sqrt{2}}} = \cdots$$

e.g.3 $B_n(R) = \{ \sum_{i=1}^n x_i^2 \le R^2 \}$ 求 $vol(B_n(R))$

$$vol(B_n(R)) = \int\cdots\int_{\sum x_i^2 \le R^2} dx_1 \cdots dx_n = \int_{-R}^{R} dx_n \int\cdots\int_{x_1^2+\cdots+x_{n-1}^2 \le R^2 - x_n^2} dx_1 \cdots dx_{n-1}$$

$$= \int_{-R}^{R} dx_n \; vol\left(B_{n-1}\left(\sqrt{R^2-x_n^2}\right)\right)$$

用归纳法建立 $vol(B_n) = C_n R^n$ $(C_1=2,\ C_2=\pi,\ C_3=\frac{4}{3}\pi)$

$$\Rightarrow C_n R^n = \int_{-R}^{R} C_{n-1}(R^2-x_n^2)^{\frac{n-1}{2}}\, dx_n \quad \underline{\underline{x_n = R\cos\theta}} \quad \int_{\pi}^{0} -C_{n-1}R^n \sin^n\theta\, d\theta$$

$$\Rightarrow C_n = \underbrace{\left(2\int_0^{\frac{\pi}{2}} \sin^n\theta\, d\theta\right)}_{=I_n,\ \text{求出}\ I_n\ \text{即可}} \cdot C_{n-1}$$

2. 换元公式

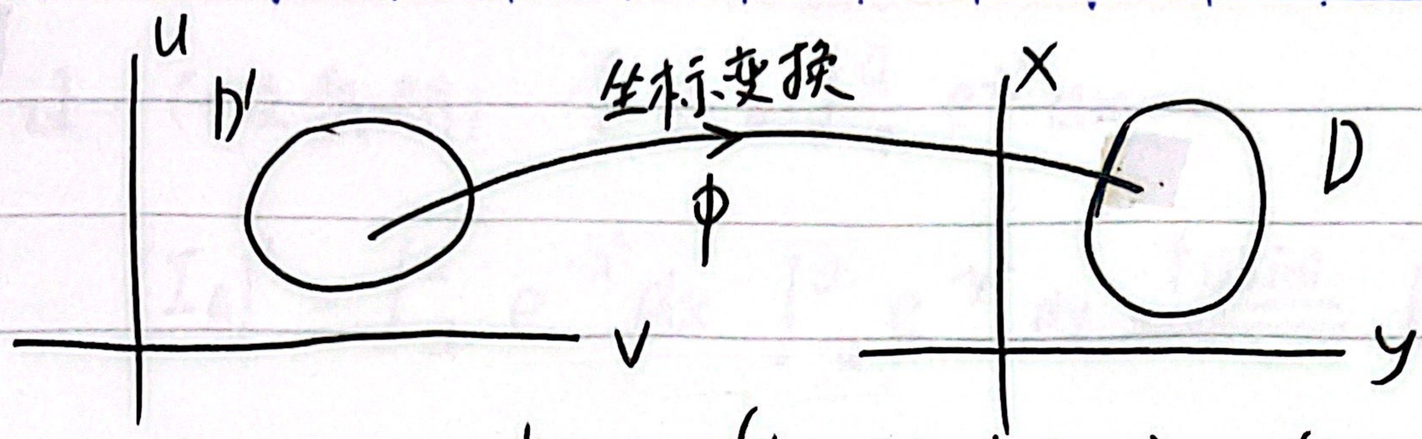
(1) Thm: 给定 $\mathbb{R}^n$ 中的 k 个向量 $\vec{v}_1, \cdots, \vec{v}_k$ $(k \le n)$，定义 $\vec{v}_1, \cdots, \vec{v}_k$ 生成的平行 k 维体为 $P = \{ \sum_{i=1}^k x_i \vec{v}_i \mid 0 \le x_i \le 1 \ \forall i \}$，则 $vol(P) = \sqrt{\det G}$ $G = \begin{pmatrix} v_1v_1 & \cdots & v_1v_k \\ \vdots & & \vdots \\ v_kv_1 & \cdots & v_kv_k \end{pmatrix}$

esp. 定义 $\mathbb{R}^3$ 中 $\vec{a}\times\vec{b} = (a_2b_3-a_3b_2,\ a_3b_1-a_1b_3,\ a_1b_2-a_2b_1)$

则 $\underline{area(\vec{a},\vec{b} \text{张成的}\square) = |\vec{a}\times\vec{b}| = \sqrt{(a_2b_3-a_3b_2)^2+(a_3b_1-a_1b_3)^2+(a_1b_2-a_2b_1)^2}}$

(Lagrange 恒等式: $(\sum a_i^2)(\sum b_i^2) - (\sum a_ib_i)^2 = \sum_{i<j}(a_ib_j-a_jb_i)^2$)

(2) 换元公式

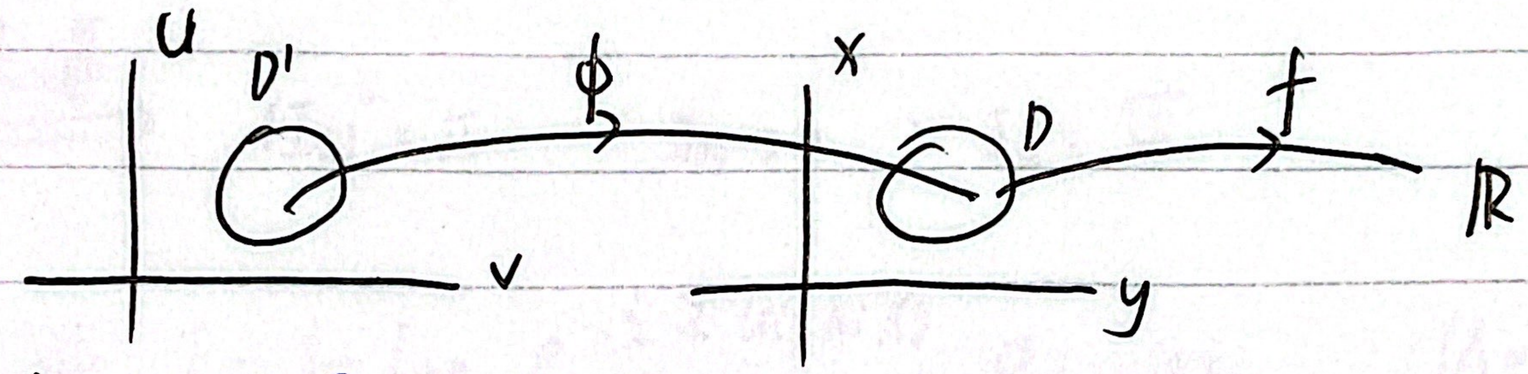


目标：$\iint_D f\,dxdy$

$$\phi(u,v)=(\phi_1(u,v),\phi_2(u,v))=(x(u,v),y(u,v))$$

Thm：设 $\phi: D'\to D$ 是 C^1 的双射，且 $\det J_\phi \neq 0$，则对 D 上可积函数 f，有

$$\iint_D f\,dxdy=\iint_{D'}(f\circ\phi)\,|\det J_\phi|\,du\,dv$$



关键：① 权重拉回：$\phi^* f \triangleq f\circ\phi$

② 坐标变换中引起的放缩倍数：$|\det J_\phi|$

n维版：$\int_D f\,|dx_1\cdots dx_n|=\int_{D'}(f\circ\phi)\,|\det J_\phi|\,|du_1\cdots du_n|$

e.g.1 $E=\left\{\sum \frac{x_i^2}{a_i^2}\le 1\right\}_{(a_i>0)}$ 求 $vol(E)$

$vol(E)=\int_E |dx_1\cdots dx_n|$ 令 $x_i=a_i u_i$ $E'=\{\sum u_i^2\le 1\}$

$$J_\phi=\begin{pmatrix} a_1 & & \\ & \ddots & \\ & & a_n\end{pmatrix} \quad |\det J_\phi|=a_1\cdots a_n$$

$$\therefore \int_E |dx_1\cdots dx_n|=\int_{E'} a_1\cdots a_n\,|du_1\cdots du_n|=a_1\cdots a_n c_n$$

e.g.2 ① 极坐标换元：$\begin{cases}x=r\cos\theta\\ y=r\sin\theta\end{cases}$ $0\le\theta\le 2\pi$ $J_\phi=\begin{pmatrix}\cos\theta & -r\sin\theta\\ \sin\theta & r\cos\theta\end{pmatrix}$

$$\Rightarrow \iint_D f(x,y)\,dxdy=\iint_{\phi^{-1}(D)} f(r\cos\theta,r\sin\theta)\,r\,dr\,d\theta$$

② 球坐标 $\begin{cases}z=r\cos\theta\\ y=r\sin\theta\sin\varphi\\ x=r\sin\theta\cos\varphi\end{cases}$ $0\le\theta\le\pi$ $0\le\varphi\le 2\pi$ $|\det J_\phi|=r^2\sin\theta$

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e.g.30 (无穷积分) (Gauss) $\int_{-\infty}^{+\infty} e^{-x^2}dx = \sqrt{\pi}$

证: (做截断) 令 $I_a = \int_{-a}^{a} e^{-x^2}dx$

$$(I_a)^2 = \int_{-a}^{a} e^{-x^2}dx \int_{-a}^{a} e^{-y^2}dy \xlongequal{\text{Fubini}} \iint_{[-a,a]^2} e^{-x^2-y^2}dxdy$$

$D_1 = \{x^2+y^2 \le a^2\} \subseteq [-a,a]^2 \subseteq D_2 = \{x^2+y^2 \le 2a^2\}$

$$\Rightarrow \iint_{D_1} \le \iint_{[-a,a]^2} \le \iint_{D_2} \quad \left(\iint_{x^2+y^2\le R^2} e^{-x^2-y^2}dxdy = \iint_{\substack{0\le r\le R\\ 0\le\theta\le 2\pi}} e^{-r^2}rdrd\theta = 2\pi\int_0^R e^{-r^2}rdr = \underbrace{\pi(1-e^{R^2})}_{\lim_{R\to\infty}=\pi}\right)$$

夹逼 $\Longrightarrow (I_a)^2 = \pi \quad \because I_a > 0 \quad \therefore I_a = \sqrt{\pi}$

② 求 $I = \int\cdots\int_{\mathbb{R}^n} e^{-\frac{1}{2}\sum a_{ij}x_ix_j}dx_1\cdots dx_n \quad (A=(a_{ij})\ 正定)$

(定义 $\int\cdots\int_{\mathbb{R}^n} f = \lim_{R\to+\infty}\int\cdots\int_{\sum_{i=1}^n x_i^2\le R^2} f$)

$$\int_{\mathbb{R}^n} e^{-\frac{1}{2}x^TAx} = \int_{\mathbb{R}^n} e^{-\frac{1}{2}x^TQ^T\Lambda Qx} \xlongequal[\phi(\vec{y}_1)=Q^T\vec{y}=Q^{-1}\vec{y}]{\vec{y}=Q\vec{x}} \int_{\mathbb{R}^n} e^{-\frac{1}{2}\sum\lambda_iy_i^2}dy_1\cdots dy_n$$

J_ϕ = ϕ的表示矩阵 $= Q^T$

$\therefore |\det J_\phi| = |\pm 1| = 1$

$$\xlongequal{\text{Fubini}} \prod_i \int_{-\infty}^{+\infty} e^{-\frac{1}{2}\lambda_iy_i^2}dy_i \xlongequal{u=\sqrt{\lambda_i}y_i} \prod_i \sqrt{\pi}\sqrt{\frac{2}{\lambda_i}} = \frac{(2\pi)^{\frac{n}{2}}}{\sqrt{\lambda_1\cdots\lambda_n}} = \frac{(2\pi)^{\frac{n}{2}}}{\sqrt{\det A}}$$

e.g.4 (正交变换的保距性)

求 $\iint_{x^2+y^2\le R^2} f(ax+by)dxdy \quad (a,b\neq 0,\ f\in C)$

考虑让 $\begin{matrix}u=ax+by\\ v=cx+dy\end{matrix}$ 即 $\begin{pmatrix}u\\v\end{pmatrix} = \begin{pmatrix}a&b\\c&d\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}$ 考虑取正交变换 $\Longrightarrow Q = \begin{pmatrix}a/\sqrt{a^2+b^2} & b/\sqrt{\ }\\ -b/\sqrt{\ } & a/\sqrt{\ }\end{pmatrix}$

① 除以 $\sqrt{a^2+b^2}$

② 令 $(c,d)\hookrightarrow(a,b)$ 逆转 90°

$$\Rightarrow \phi(u,v) = Q^{-1}\begin{pmatrix}u\\v\end{pmatrix} = Q^T\begin{pmatrix}u\\v\end{pmatrix} \quad \underline{\therefore |\det J_\phi| = 1}$$

如此1

$$\Rightarrow \iint_{x^2+y^2\le R^2} f(ax+by)dxdy = \iint_{\underline{u^2+v^2\le R^2}} f(\sqrt{a^2+b^2}u)dudv = \int_{-R}^{R} f(\sqrt{\ }u)du\left(\int_{-\sqrt{R^2-u^2}}^{+\sqrt{\ }} dv\right.$$

如此2 (保距)

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e.g.5 (用于处理对称性) (it seems obvious 而且后来用的时候也没从换元公式上思考过，but 现在写一遍吧)

命题：设 Ω 关于 oxy 平面的反射不变，则

① $f(x,y,z)$ 关于 z 奇 $\Rightarrow \iiint_\Omega f = 0$

② 偶 $\Rightarrow \iiint_\Omega f = 2\iiint_{\Omega\cap\{z\geq 0\}} f$

证：$\int_{\Omega^-} f(x,y,z)|dxdydz| \xrightarrow[\phi(x,y,z)=(x,y,-z)]{\phi:\Omega^+\to\Omega^-,\ |\det J_\phi|=|-1|=1} \int_{\Omega^+} \underbrace{(f\circ\phi)}_{\text{"}f(x,y,-z)}|dxdydz|$

$$= \int_{\Omega^+} \begin{matrix} f(x,y,z) & ② \\ -f(x,y,z) & ① \end{matrix} \quad \checkmark$$

e.g.6 "排序轮换对称性" (名我自己起的)

$V = \{(x_1,\cdots,x_n) \mid 0\leq x_1\leq\cdots\leq x_n\leq 1\}$ 求 $vol(V)$

法一) Fubini

$$\int\cdots\int_V dx_1\cdots dx_n = \int\cdots\int_{0\leq x_1\leq\cdots\leq x_{n-1}\leq 1} dx_1\cdots dx_{n-1} \underbrace{\int_{x_{n-1}}^1 dx_n}_{=(1-x_{n-1})}$$

$$= \int\cdots\int_{0\leq x_1\leq\cdots\leq x_{n-2}\leq 1} \underbrace{\int_{x_{n-2}}^1 (1-x_{n-1})\,dx_{n-1}}_{=\frac{1}{2}(1-x_{n-2})^2} = \cdots = \frac{1}{n!}$$

法二) $\int\cdots\int_{0\leq x_1\leq\cdots\leq x_n} dvol = \frac{1}{n!}\int\cdots\int_{[0,1]^n} dvol \overset{\text{Fubini}}{=\!=} \frac{1}{n!}\left(\int_0^1 dx_1\right)\cdots\left(\int_0^1 dx_n\right) = \frac{1}{n!}$

($\{x_1,\cdots,x_n\}$ 有 $n!$ 种排序 计证为 every $Vol(V_\sigma) = vol(V)$ 定义 $V_\sigma = \{x_{\sigma(1)}\leq\cdots\leq x_{\sigma(n)}\}$ (σ 取遍所有置换 $\sigma:\{1,\cdots,n\}\to\{1,\cdots,n\}$)

则 ✓ ∴下面证 $vol(V_\sigma) = vol(V)$

$\phi: V_\sigma \to V$ J_ϕ = 置换矩阵 $\therefore |\det J_\phi| = 1$

$\Rightarrow \int_V dvol = \int_{V_\sigma} dvol$ ✓

三. 曲线/曲面积分 < 第一型: ∫标量 (1/2重积分 ↔ 带权的 Length/area)
                  第二型: ∫矢量

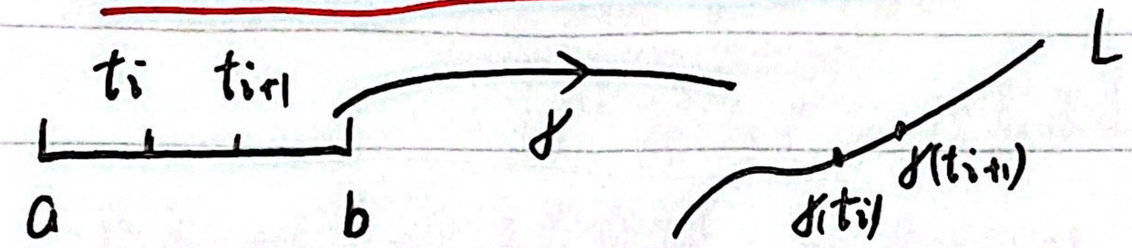
1. 第一型曲线积分

定义: 设 $L \subseteq \mathbb{R}^n$ 是光滑曲线, 权重 $f: L \to \mathbb{R}$, 求 L 的 带权长度 $\int_L f\,dl$

Solution: 参数化 $\gamma: [a,b] \to L$ s.t. ① γ 几乎单射 ($\gamma(t_1)=\gamma(t_2) \Leftrightarrow t_1=t_2$ / $\{t_1,t_2\}=\{a,b\}$)
② $\gamma'(t) \neq 0 \ \forall t$

⇒ 定义第一型曲线积分

$$\int_L f\,dl = \int_a^b f(\gamma(t))\,|\gamma'(t)|\,dt$$



($dl \sim |\gamma'(t)|\,dt$
长度 = 速度 × 时间)

esp. ① $L \subseteq \mathbb{R}^3$ $\gamma(t) = (x(t), y(t), z(t))$ $(a \le t \le b)$.

$$\int_L f(x,y,z)\,dl = \int_a^b f(x(t), y(t), z(t)) \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2}\,dt$$

(速度)

② L 极坐标方程 $r = r(\theta)$ $(\alpha \le \theta \le \beta)$

自动有参数化 $\begin{cases} x = r(\theta)\cos\theta \\ y = r(\theta)\sin\theta \end{cases}$ $\Rightarrow \int_L f\,dl = \int_\alpha^\beta f(r(\theta)\cos\theta, r(\theta)\sin\theta)\sqrt{x'(\theta)^2 + y'(\theta)^2}\,d\theta$

$$= \int_\alpha^\beta f(r(\theta)\cos\theta, r(\theta)\sin\theta)\sqrt{r'(\theta)^2 + r(\theta)^2}\,d\theta$$

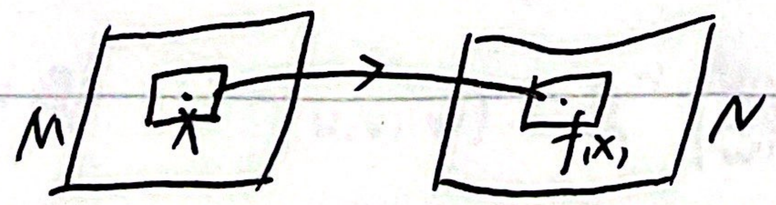
命题: 曲线长 ≥ 线段长

证明关键: $\int_a^b \gamma'(t)(\gamma(b)-\gamma(a))\,dt \le \int_a^b |\gamma'(t)|\,|\gamma(b)-\gamma(a)|\,dt = |\gamma(b)-\gamma(a)| \int_a^b |\gamma'(t)|\,dt = |AB| \cdot \text{Length}$

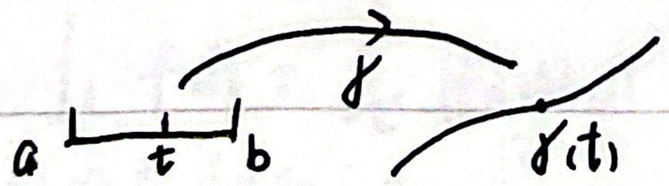
$\int_a^b \sum \gamma_i'(t)(\gamma_i(b)-\gamma_i(a))\,dt \overset{\sim L}{=\!=} \sum (\gamma_i(b)-\gamma_i(a))(\gamma_i(b)-\gamma_i(a)) = |AB|^2$

(2) 第一型曲线积分的换元公式

大致推导如下:



$M \xrightarrow{f} N$ 诱导线性映射 $f_*: T_xM \to T_{f(x)}N$



设 $\mathbb{R}^n$ 中单位正方向为 $\frac{\partial}{\partial t} \in T_t\mathbb{R}$, 则有 $\gamma(t) = \gamma_*(\frac{\partial}{\partial t})$

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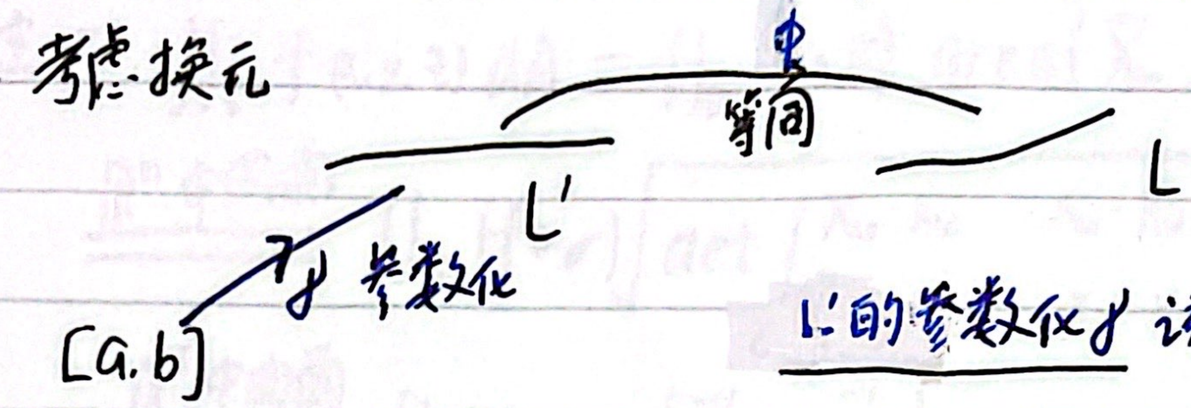
从而 $\int_L f dl = \int_a^b (f\circ\gamma)\,|\gamma_*(\frac{\partial}{\partial t})|\,dt$

考虑换元



L'的参数化 γ 诱导了 L 的参数化 $\phi\circ\gamma$

$\Rightarrow \int_L f dl = \int_a^b (f\circ\phi\circ\gamma)\,|(\phi\circ\gamma)_*(\frac{\partial}{\partial t})|\,dt$

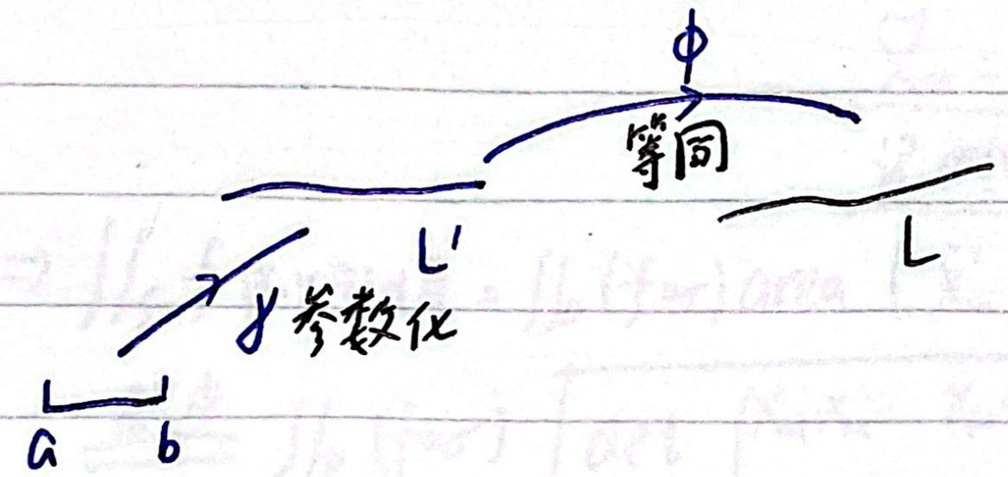
$= \phi_*(\gamma_*(\frac{\partial}{\partial t}))$，$\phi_*$ 的矩阵表示是 J_ϕ

$= \int_a^b (f\circ\phi\circ\gamma)\cdot |J_\phi(\gamma(t))\cdot\gamma'(t)|\,dt$

$= \sqrt{\gamma'^T J_\phi^T J_\phi \gamma'}$　J_ϕ 正交，则 $= \sqrt{\gamma'^T\cdot\gamma}$

$\Rightarrow$ Thm：设 $\phi:\mathbb{R}^3\to\mathbb{R}^3$ 正交变换，且 $\phi(L')=L$，则

$$\int_L f dl = \int_a^b (f\circ\phi\circ\gamma)\,|\gamma'(t)|\,dt = \int_{L'} f\circ\phi\, dl$$



（这部分似乎没啥例题？
但至少可为第二型曲面积分做铺垫）

例：$L=\{x^2+y^2+z^2=1,\ x+y+z=1\}$　求 $\int_L f dl$

（考虑换元让 $w=x+y+z=1$）

令 $\begin{pmatrix}u\\v\\w\end{pmatrix} = \begin{pmatrix} & & \\ & & \\ \frac{1}{\sqrt3} & \frac{1}{\sqrt3} & \frac{1}{\sqrt3}\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}$

Q正交

（how求Q？→ Q性质：每行列向量模长=1，每行列向量相互⊥）

∴可取 $q_1^T=(1/\sqrt2,-1/\sqrt2,0)$

$q_2^T=(1/\sqrt6,\ 1/\sqrt6,\ -2/\sqrt6)$

$\Rightarrow \phi:\mathbb{R}^3\to\mathbb{R}^3$　$\phi(u,v,w)=Q^{-1}\begin{pmatrix}u\\v\\w\end{pmatrix}=Q^T\begin{pmatrix}u\\v\\w\end{pmatrix}=\begin{pmatrix}x\\y\\z\end{pmatrix}$

$\phi^{-1}(L)=L'=\{u^2+v^2+w^2=1,\ w=1/\sqrt3\}$　∵保距

$\Rightarrow \int_L f dl = \int_{L'} f\circ\phi\, dl$

L'有参数化 $\begin{cases}u=\frac{\sqrt6}{3}\cos t\\ v=\frac{\sqrt6}{3}\sin t\\ w=\frac{1}{\sqrt3}\end{cases}$ $(0\le t\le 2\pi)$

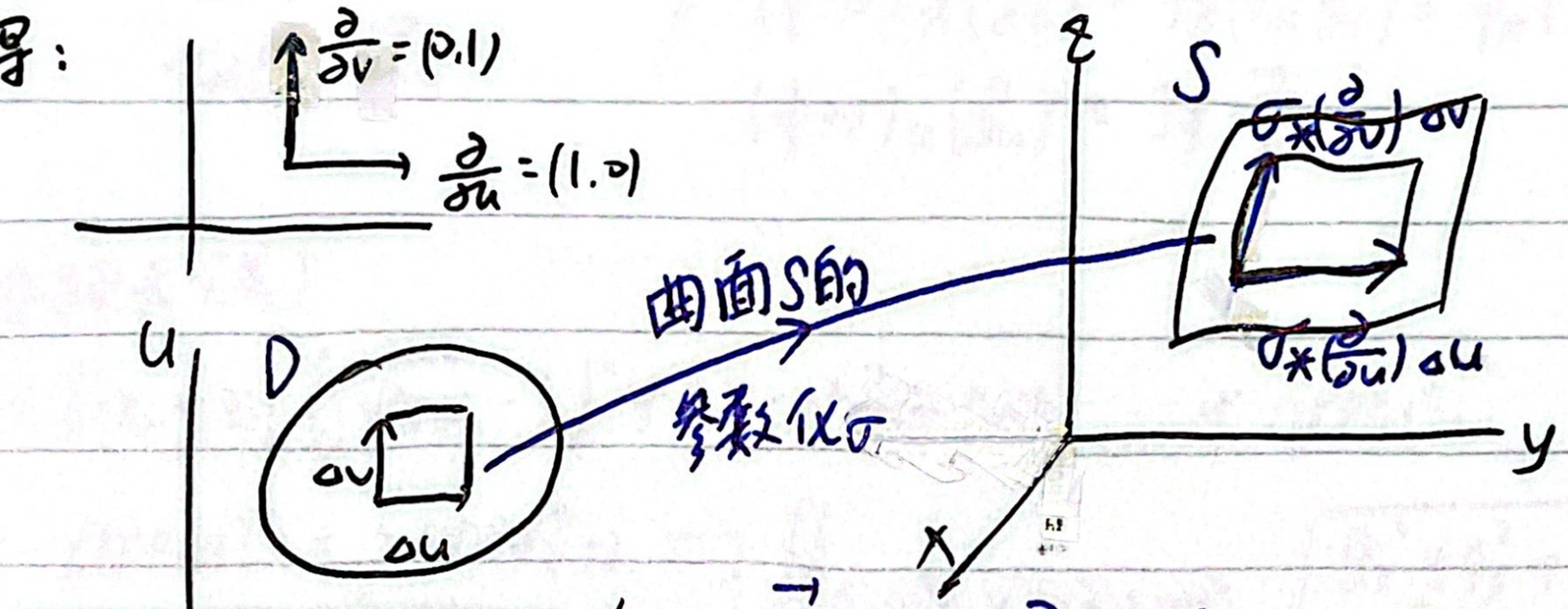
2. 第一型曲面积分

定义 $\iint_S f(x,y,z)\,dA = \iint_D (f\circ\sigma)\,\text{area}(\vec{X}_u, \vec{X}_v \text{张成的}\square)\,du\,dv$

$\underline{\mathbb{R}^n\text{中曲面}}$ $\iint_D (f\circ\sigma)\sqrt{\det\begin{pmatrix} X_u\cdot X_u & X_u\cdot X_v \\ X_v\cdot X_u & X_v\cdot X_v\end{pmatrix}}\,du\,dv$

$\underline{\mathbb{R}^3\text{中曲面}}$ $\iint_D (f\circ\sigma)\,|\vec{X}_u\times\vec{X}_v|\,du\,dv$

大致推导：



$\vec{X}_u = \sigma_*(\frac{\partial}{\partial u}) = (x_u, y_u, z_u)$

$\vec{X}_v = \sigma_*(\frac{\partial}{\partial v}) = (x_v, y_v, z_v)$

记 $\sigma(u,v) = (x(u,v),\ y(u,v),\ z(u,v))$

$\Rightarrow \iint_S f(x,y,z)\,dA = \iint_D (f\circ\sigma)\,\text{area}(\vec{X}_u, \vec{X}_v \text{张成的}\square)\,du\,dv$

$\underline{\mathbb{R}^n\text{中}}$ $\iint_D (f\circ\sigma)\sqrt{\det\begin{pmatrix} X_u\cdot X_u & X_u\cdot X_v \\ X_v\cdot X_u & X_v\cdot X_v\end{pmatrix}}\,du\,dv$

esp. if S 可视为 $g: D\to\mathbb{R}$ 的图像，即 $S = \text{graph}(g) = \{(x,y,z)\mid (x,y)\in D\}$

$\Rightarrow S$ 有参数化 $\begin{cases} x = u \\ y = v \\ z = g(x,y)\end{cases}$ $\Rightarrow \iint_S f(x,y,z)\,dA = \iint_D f(x,y,g(x,y))\sqrt{g_x^2+g_y^2+1}\,dx\,dy$

e.g.1 $S^2 = \{x^2+y^2+z^2 = R^2\}$ 求 $\text{Area}(S^2)$

法一) $\text{Area}(S^2) = 2\,\text{Area}(S_+^2)$ $\underline{\underline{S_+^2\text{视为 graph}\left(g=\sqrt{R^2-x^2-y^2}\right)}}$ $2\iint_{x^2+y^2\le R^2}\sqrt{g_x^2+g_y^2+1}\,dx\,dy$

$= 2\iint_{x^2+y^2\le R^2}\sqrt{\frac{R^2}{R^2-x^2-y^2}}\,dx\,dy = 2R\iint_{\substack{0\le r\le R\\ 0\le\theta\le 2\pi}}\frac{1}{\sqrt{R^2-r^2}}\,r\,dr = 4\pi R\left(-\sqrt{R^2-r^2}\right)\Big|_0^R = 4\pi R^2$

法二) 直接参数化 $(\sigma = \sigma(\theta,\varphi))$ $\begin{cases} z = R\cos\theta \\ y = R\sin\theta\sin\varphi \\ x = R\sin\theta\cos\varphi\end{cases}$

$\vec{X}_\theta = (R\cos\theta\cos\varphi,\ R\cos\theta\sin\varphi,\ -R\sin\theta)$

$\vec{X}_\varphi = (-R\sin\theta\sin\varphi,\ R\sin\theta\cos\varphi,\ 0)$

$|\vec{X}_\theta\times\vec{X}_\varphi| = R^2\sqrt{s^4\theta c^2\varphi + s^4\theta s^2\varphi + s^2\theta c^2\theta} = R^2\sin\theta$ ✓

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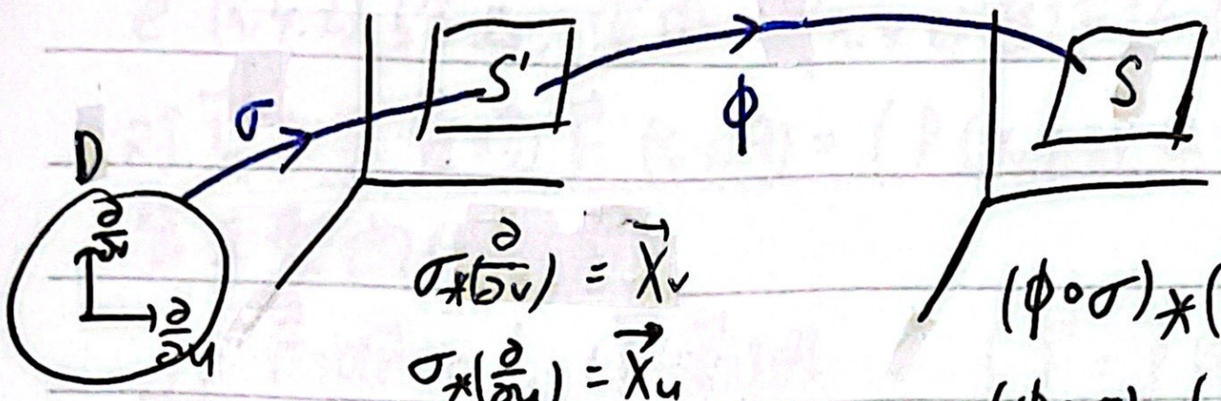
e.g.2 (Co-area formula) 略

(2) 第I型曲面积分换元公式

Thm: 设 $\phi: \mathbb{R}^3 \to \mathbb{R}^3$ 正交, 则 $\iint_S f\,dA = \iint_{S'} (f\circ\phi)\,dA$



$\sigma_*(\frac{\partial}{\partial v}) = \vec{X}_v$

$\sigma_*(\frac{\partial}{\partial u}) = \vec{X}_u$

$(\phi\circ\sigma)_*(\frac{\partial}{\partial v}) = \phi_*(\sigma_*\frac{\partial}{\partial v}) = \phi_*(\vec{X}_v) = J_\phi\cdot\vec{X}_v$

$(\phi\circ\sigma)_*(\frac{\partial}{\partial u}) = J_\phi\cdot\vec{X}_u$

e.g: (非换元公式)

$T = \{(x,y,z) \mid (\sqrt{x^2+y^2}-b)^2 + z^2 = a^2\}$ $(a<b)$, 求 area(T)

法一 $\text{area}(T) = 2\,\text{area}(T_+) = 2\iint_{(b-a)^2 \le x^2+y^2 \le (b+a)^2} \sqrt{g_x^2+g_y^2+1}\,dx\,dy$

法二) 直接参变元化

$\sigma = \sigma(\alpha,\beta)$

$\vec{X}_\alpha = \cdots \quad \vec{X}_\beta = \cdots$

$$\begin{cases} z = a\cos\beta \\ y = (a\sin\beta + b)\sin\alpha \\ x = (a\sin\beta + b)\cos\alpha \end{cases} \qquad 0 \le \alpha, \beta \le 2\pi$$

$$\text{area}(T) = \iint_{\substack{0\le\alpha\le 2\pi \\ 0\le\beta\le 2\pi}} |\vec{X}_\alpha \times \vec{X}_\beta|\,d\alpha\,d\beta$$

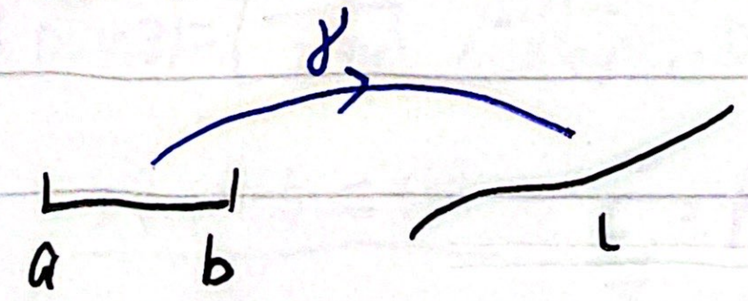
3. 第二型曲线积分 (第二型 ↔ 矢量场的积分)

设矢量场 $\vec{F} = (P,Q,R): L \to \mathbb{R}^3$, 定义 $\vec{F}$ 在 L 上的第II型曲线积分为

$$\int_L \vec{F}\cdot d\vec{r} = \int_a^b \left(P(x(t),y(t),z(t))\,x'(t) + Q\,y'(t) + R\,z'(t)\right)dt \quad (*1)$$

先不看了这么写有点

($x(t), y(t), z(t)$ 源于参数化 γ:)



$\gamma(t) = (x(t), y(t), z(t))$

$(x'(t), y'(t), z'(t))$ 为速度矢量,

$d\vec{r} = (x'(t), y'(t), z'(t))\,dt$

形式化 $x'(t)\,dt = dx$ → 定义: 记 $\int_L (P,Q,R)\,d\vec{r} = \int_L P\,dx + Q\,dy + R\,dz$ (第II型曲线积分) (*2)

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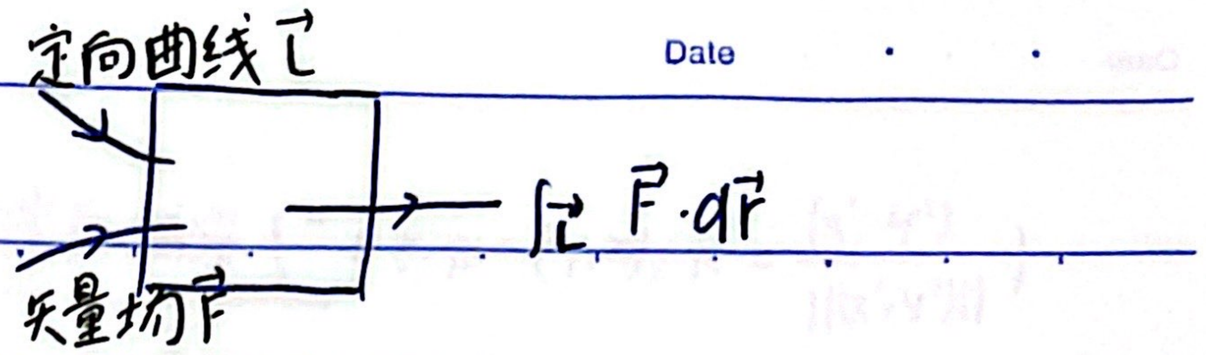
重新梳理框架：第二型曲线积分



设 $\vec{L}$ 是定向曲线，其定向由在 $\vec{L}$ 上每点 (x,y,z) 给 L 指定一个单位切向量 $\vec{e}(x,y,z)$ 给出，要求 $\vec{e}(x,y,z)$ 连续依赖于点 (x,y,z)，即 $\vec{e}: L \to \mathbb{R}^3 \in C$（相容性条件）

对 $\vec{L}$ 上矢量场 $\vec{F}(x,y,z) = (P(x,y,z), Q(x,y,z), R(x,y,z))$ 定义 $\vec{F}$ 在 $\vec{L}$ 上的第二型曲线积分

$$\int_{\vec{L}} \vec{F}\cdot d\vec{r} = \int_L (\vec{F}\cdot\vec{e})\,dl \qquad (\vec{F}=\vec{F}(x,y,z),\ \vec{e}=\vec{e}(x,y,z))$$

计算方法：选与定向相容的参数化 $\gamma: [a,b] \to L$（计 $\vec{e}(\gamma(t)) = \frac{\gamma'(t)}{|\gamma'(t)|}$，则称 γ 与定向相容）

则有：$\int_{\vec{L}} \vec{F}\cdot d\vec{r} = \int_L (\vec{F}\cdot\vec{e})\,dl = \int_a^b \left(\vec{F}\cdot\frac{\gamma'(t)}{|\gamma'(t)|}\right)|\gamma'(t)|\,dt$

$$= \int_a^b \vec{F}(\gamma(t))\cdot\gamma'(t)\,dt = \int_a^b \big(P(x(t),y(t),z(t))x'(t) + Q\,y'(t) + R\,z'(t)\big)\,dt$$

由此，也记：$\int_{\vec{L}} \vec{F}\,d\vec{r} = \int_{\vec{L}} P\,dx + Q\,dy + R\,dz$

e.g.1 命题：设 $\vec{F} = \nabla\phi$（ϕ 势能，$\phi \in C^1$），则 $\int_{\vec{L}} \vec{F}\cdot d\vec{r} = \phi(L\text{终点}) - \phi(L\text{起点})$

证：选与定向相容的 γ，$\vec{L}$ 起点 $= \gamma(a)$，$\vec{L}$ 终点 $= \gamma(b)$

$$\int_{\vec{L}} \vec{F}\cdot d\vec{r} = \int_a^b \left(\frac{\partial\phi}{\partial x}x'(t) + \frac{\partial\phi}{\partial y}y'(t) + \frac{\partial\phi}{\partial z}z'(t)\right)dt$$

$$= \int_a^b \left(\frac{d}{dt}\phi(x(t),y(t),z(t))\right)dt = \phi(\gamma(t))\Big|_{t=a}^{t=b} = \phi(\gamma(b)) - \phi(\gamma(a)) \quad \checkmark$$

推论：$\int_{\vec{L}\text{封闭}} \nabla\phi\cdot d\vec{r} = 0$

（三元加看道回指十指向）

e.g.2 (Gauss) $\vec{B}(x,y,z) = \left(\frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2}, 0\right)$，$C = \{x^2+y^2=R^2 \text{ 且 } z=0\}$，求 $\int_{\vec{C}} \vec{B}\cdot d\vec{r}$

法一 $I = \int_{\vec{C}} \vec{B}\cdot d\vec{r}$　约化为二维：$\tilde{C} = \{(x,y) \mid x^2+y^2=R^2\}$　$\vec{e}$ 逆时针定向

$\vec{e}(x,y) = (-y,x)/R$

$\vec{\tilde{B}} = (\ ,\ )$

$$\Rightarrow I = \int_{\vec{C}} \vec{\tilde{B}}\,d\vec{r} = \int_{\tilde{C}} (\vec{\tilde{B}}\cdot\vec{e})\,dl = \int_{\tilde{C}} \left(\frac{-y}{R^2}, \frac{x}{R^2}\right)\cdot\left(\frac{-y}{R}, \frac{x}{R}\right)dl$$

$$= \int_{\tilde{C}} \frac{1}{R}\,dl = 2\pi$$

（第一型曲线积分）

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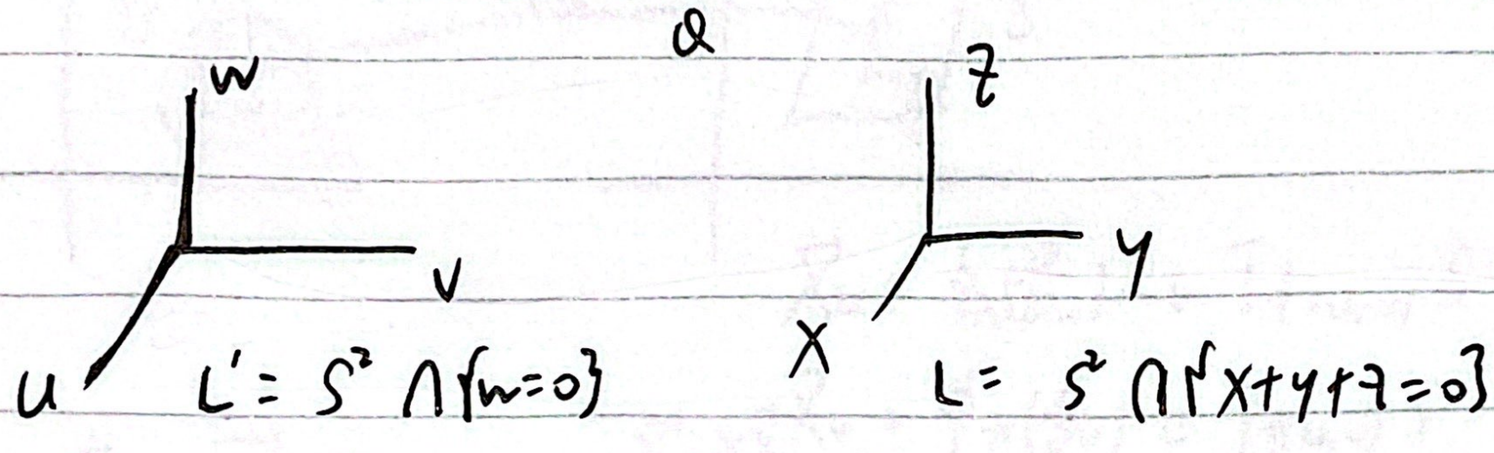
法二) 直接参数化 $\begin{cases} x = R\cos t \\ y = R\sin t \end{cases} (0 \le t \le 2\pi)$ 与定向相容 ($\because \vec{e}_{(x,y)} = (-y,x)/R = \frac{(x',y')}{\|(x',y')\|}$)

$$\Rightarrow I = \int_{\vec{C}} \frac{-y\,dx + x\,dy}{R^2} = \int_0^{2\pi} \frac{-R\sin t\,(-R\sin t) + R\cos t \cdot R\cos t}{R^2}\,dt = 2\pi$$

e.9.3 $\vec{L} = \{x^2+y^2+z^2=1,\ x+y+z=0\}$ 定向：$z=+\infty$ 看为逆，求 $\int_{\vec{L}} z\,dx + x\,dy + y\,dz$

写参数化，验证相容：

令 $\begin{pmatrix} u \\ v \\ w \end{pmatrix} = \underbrace{\begin{pmatrix} & & \\ & & \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}}_{Q} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ $Q = \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{6} & 1/\sqrt{6} & -2/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{pmatrix}$ （Q为特殊正交变换，即 $QQ^T = I_3$, $\det Q = 1$，物理意义：旋转）



在 (u, v, w) 坐标下找与 L' 定向相容的参数化

$\begin{cases} u(t) = \cos t \\ v(t) = \sin t \\ w(t) = 0 \end{cases} (0 \le t \le 2\pi)$ —诱导L的与L定向相容的参数化→ $\begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix} = Q^T \begin{pmatrix} \cos t \\ \sin t \\ 0 \end{pmatrix}$

（以上过程只是为了找 (x,y,z) 的参数化，思路与I型曲线/曲面积分的换元法很像）

$$\Rightarrow \int_{\vec{C}} z\,dx + x\,dy + y\,dz = \int_0^{2\pi} \cdots dt = \cdots$$

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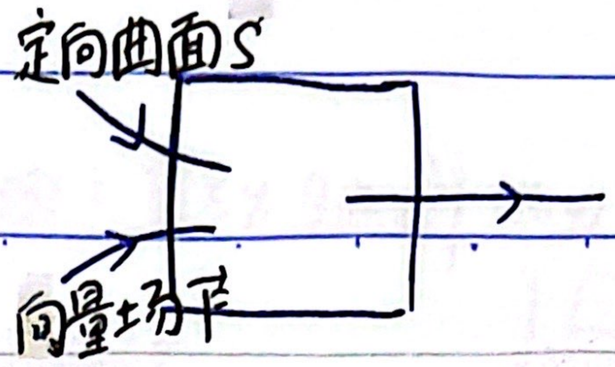
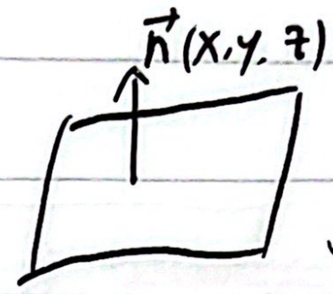
4. 第二型曲面积分



$\iint_S \vec{F}\cdot d\vec{s}$ (流体流量)

设 S 是定向曲面，其定向由在 S 上每点 (x,y,z) 处指定 S 的一个单位法向量 $\vec{n}(x,y,z)$ 给出，要求 $\vec{n}(x,y,z)$ 连续依赖于点 (x,y,z)



(通常默认 $\vec{n}(x,y,z)$ 指向 S 外侧/上方)

对 S 上的向量场 $\vec{F}(x,y,z)=(P(x,y,z),Q(x,y,z),R(x,y,z))$

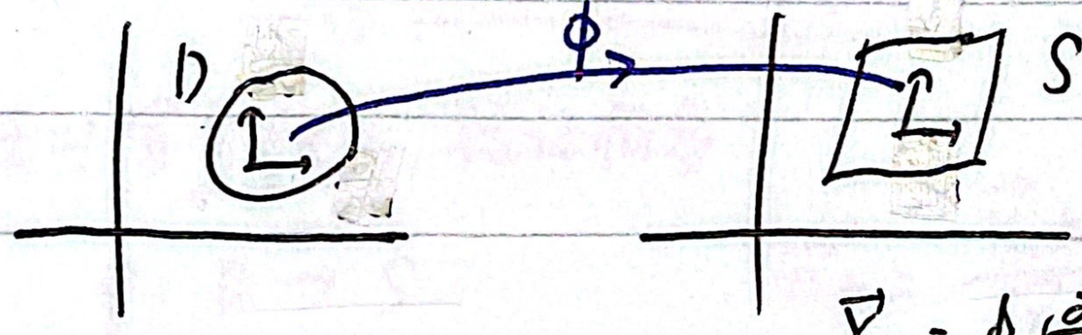
定义 $\vec{F}$ 在 S 上的第二型曲面积分为：

$$\iint_S \vec{F}\cdot d\vec{S} = \iint_S (\vec{F}\cdot\vec{n})\,dS$$

第二型曲线积分：$\vec{F}$ 点乘 $\vec{e}$ 转化为二型曲线

二 曲面：$\vec{F}$ $\vec{n}$ 二型曲面

命题，参数化诱导定向



$\vec{X}_u = \phi_*(\frac{\partial}{\partial u}) \in T_{\phi(u,v)}S$

$\vec{X}_v = \phi_*(\frac{\partial}{\partial v}) \in T_{\phi(u,v)}S$

定义 $\vec{n}_\phi(\phi(u,v)) = \dfrac{\vec{X}_u \times \vec{X}_v}{|\vec{X}_u \times \vec{X}_v|}$ 称为 ϕ 诱导的定向

$\dfrac{\vec{X}_u \times \vec{X}_v}{|\vec{X}_u \times \vec{X}_v|} = \vec{n}$

计算方法：选一个与定向相容的参数化 ϕ ($\Leftrightarrow \vec{n}_\phi = \vec{n}$ 即 $\vec{X}_u \times \vec{X}_v \propto \vec{n}$)

因此，$\iint_S \vec{F}\cdot d\vec{S} = \iint_S (\vec{F}\cdot\vec{n})\,dS \xrightarrow[\text{算}]{\text{用}\phi} \iint_D \vec{F}(\phi(u,v))\cdot\vec{n}_\phi(\phi(u,v))\cdot|\vec{X}_u\times\vec{X}_v|\,du\,dv$

$$= \iint_D \vec{F}(\phi(u,v))\cdot(\vec{X}_u\times\vec{X}_v)\,du\,dv$$

$$= \iint_D \det\begin{pmatrix} P & Q & R \\ x_u & y_u & z_u \\ x_v & y_v & z_v \end{pmatrix} du\,dv$$

(注：$\vec{a}\cdot(\vec{b}\times\vec{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$)

展开得到 $\iint_D P(x(u,v),y(u,v),z(u,v))\,(y_u z_v - y_v z_u)\,du\,dv + \text{etc}$

$\sim dy\,dz$

由此，记 $\iint_S \vec{F}\cdot d\vec{S} = \iint_S P\,dy\,dz + Q\,dz\,dx + R\,dx\,dy$

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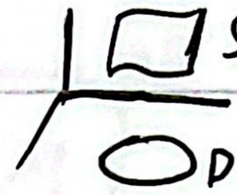
e.g.1 命题 $S = \text{graph}(g) = \{(x,y,g(x,y)) \mid (x,y) \in D\}$，则 $\iint_S \vec{F}\cdot d\vec{S} = \iint_D \vec{F}(-g_x,-g_y,1)\,dx$



法一) 求出 $\vec{n} = \dfrac{(-g_x,-g_y,1)}{\sqrt{g_x^2+g_y^2+1}}$

$$\Rightarrow \iint_S \vec{F}\cdot d\vec{S} = \iint_S \vec{F}\cdot\vec{n}\,dS = \iint_S \vec{F}\cdot(-g_x,-g_y,1)\frac{1}{\sqrt{g_x^2+g_y^2+1}}\,dS$$

$$= \iint_D \vec{F}(-g_x,-g_y,1)\frac{1}{\sqrt{g_x^2+g_y^2+1}}\sqrt{g_x^2+g_y^2+1}\,dx\,dy \checkmark$$

(现在用第1型曲面积分的方法就ok啦!)

how 求 $\vec{n}$? $\text{graph}(g) = \{f(x,y,z) = z - g(x,y) = 0\}$

~~法方向 = span{∇f}~~ $\therefore \vec{n} = \dfrac{(-g_x,-g_y,1)}{\sqrt{g_x^2+g_y^2+1}}$

法二) 写参数化，验证相容

$\phi: D \to S$

$(x,y) \to (x,y,g(x,y))$

$$\vec{n}_\phi = \frac{\vec{X}_u \times \vec{X}_v}{|\vec{X}_u \times \vec{X}_v|} = \frac{(1,0,g_x)\times(0,1,g_y)}{1} = \frac{(-g_x,-g_y,1)}{\sqrt{}}$$ 指向S上方 ∴相容

$$\Rightarrow \iint_S \vec{F}\cdot d\vec{S} \overset{\text{用}\phi\text{算}}{=\!=} \iint_D \vec{F}\cdot(\vec{X}_u\times\vec{X}_v)\,du\,dv = \iint_D \vec{F}\cdot(-g_x,-g_y,1)\,dx\,dy$$

e.g.2 (Gauss) $\vec{E}(x,y,z) = \dfrac{(x,y,z)}{(x^2+y^2+z^2)^{\frac{3}{2}}}$ 求 $\iint_S \vec{E}\cdot d\vec{S}$ $S = \{x^2+y^2+z^2 = R^2\}$

$$\iint_S \vec{E}\cdot d\vec{S} = \iint_S \vec{E}\cdot\vec{n}\cdot dS = \iint_S \frac{(x,y,z)}{R^2}\cdot\frac{(x,y,z)}{R}\,dS = \frac{1}{R^2}\iint_S dS = 4\pi$$

$\vec{n} = \dfrac{(x,y,z)}{R}$

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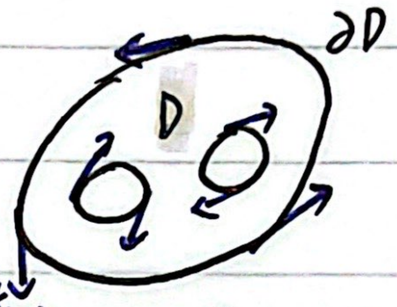
四. Green & Stokes

1. Green 公式

对平面区域 D 的边界 ∂D 赋予

边界正定向 ∂D^+：沿 ∂D 正定向前进



D 总在人的左手边 (∂D^+ $\begin{cases}\text{外边界 } C_0: \text{逆}\\ \text{洞 } C_i: \text{顺}\end{cases}$)，对 D 上的 C^1 光滑※函数 P, Q 有

$$\int_{\partial D^+} (P\,dx + Q\,dy) = \iint_D \left(\frac{\partial P}{\partial y} + \frac{\partial Q}{\partial x}\right) dx\,dy \qquad (D \subseteq \mathbb{R}^2,\ P = P(x,y),\ Q = Q(x,y))$$

e.g.1 可选 P, Q 使 RHS = area(D)

① $P = 0,\ Q = x$ $\quad \int_{\partial D} x\,dy = \text{area}(D)$ $\quad$ (∂D^+ 的“十”略)

② $P = -y,\ Q = 0$ $\quad \int_{\partial D} -y\,dx = \text{area}(D)$

③ $P = -\frac{y}{2},\ Q = \frac{x}{2}$ $\quad \int_{\partial D} -\frac{y}{2}dx + \frac{x}{2}dy = \text{area}(D)$

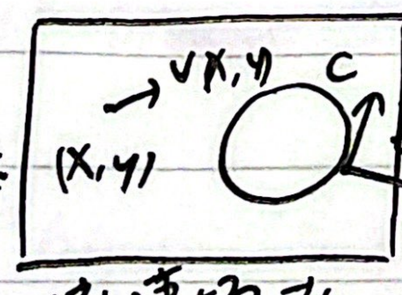
e.g.2 求 area ($D = \left\{\frac{x^2}{a^2} + \frac{y^2}{b^2} \le 1\right\}$) $(a, b > 0)$

由 e.g.1 ③，$\text{area}(D) = \int_{\partial D} -\frac{y}{2}dx + \frac{x}{2}dy \xrightarrow[\substack{x = a\cos\theta\\ y = b\sin\theta}]{\partial E \text{ 参数化}} \int_0^{2\pi} \frac{ab}{2}\,d\theta = \pi ab$ (与定向相容)

不要忘记心里验证一下

e.g.3 (Green 平面流体流量公式)

平面流体



流速场 $\vec{v}(x,y) = (v_1, v_2)$

求 $I = \int_C \vec{v}(x,y) \cdot \vec{n}(x,y)\,dl$

$$I = \int_C \vec{v}(x,y) \cdot \vec{n}(x,y)\,dl \xrightarrow[\text{保内积}]{\text{正交变换}} \int_C \underbrace{(-v_2, v_1)}_{\vec{v} \text{ 逆转 } 90^\circ} \cdot \underbrace{\vec{e}}_{\vec{n} \text{ 逆转 } 90^\circ} dl = \int_C -v_2\,dx + v_1\,dy \quad (\text{正型曲线})$$

Green $\iint_D \left(-\frac{\partial(-v_2)}{\partial y} + \frac{\partial v_1}{\partial x}\right) dx\,dy = \iint_D \left(\frac{\partial v_2}{\partial y} + \frac{\partial v_1}{\partial x}\right) dx\,dy$

esp. 对 $\vec{v} = \nabla u = (u_x, u_y)$ (有势场，u 为势能)，则

$$I = \int_C (\vec{v} \cdot \vec{n})\,dl = \int_C (\nabla \vec{u} \cdot \vec{n})\,dl = \int_C \frac{\partial u}{\partial \vec{n}}\,dl = \iint_D (\Delta u)\,dx\,dy$$

(回忆：$\nabla f \cdot \vec{v} = \underbrace{\nabla_{\vec{v}} f}_{\text{方向导数}}$)

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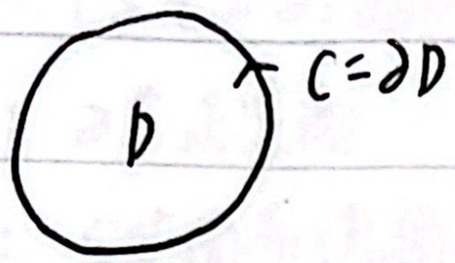
e.g.4 设 C 为简单闭曲线 ($C \subseteq \mathbb{R}^2$), 求 $I = \int_{C逆} \frac{-y\,dx + x\,dy}{x^2+y^2}$



① $\vec{0} \notin D$ 即 $\vec{0}$ 在 C 外

$\Rightarrow P, Q \in C^1(D)$ $\therefore \int_{\partial D^+} \frac{-y\,dx + x\,dy}{x^2+y^2} \overset{Green}{=\!=} \iint_D \left[-\frac{\partial}{\partial y}\left(\frac{-y}{x^2+y^2}\right) + \frac{\partial}{\partial x}\left(\frac{x}{x^2+y^2}\right)\right] dx\,dy = 0$

② $\vec{0} \in D$

定义 $E = D \setminus \{x^2+y^2 < \varepsilon^2\}$, 则 $P, Q \in C^1(E)$

$$\int_{\partial E^+} P\,dx + Q\,dy = \iint_E -\frac{\partial P}{\partial y} + \frac{\partial Q}{\partial x} = 0$$

$$\int_C + \int_{x^2+y^2=\varepsilon^2 顺} = 0$$

$\Rightarrow I = \int_C = \int_{x^2+y^2=\varepsilon^2 逆} (P\,dx + Q\,dy) = 2\pi$

综上, $\int_C \frac{-y\,dx + x\,dy}{x^2+y^2} = \begin{cases} 0 & \Leftrightarrow \vec{0} 在 C 外 \\ 2\pi & \Leftrightarrow \vec{0} 在 C 内 \end{cases}$

e.g.5 Brouwer 不动点 Thm 略

2. Gauss 公式 (Green 推广至三维)

对有界三维区域 Ω 的边界 $\partial\Omega$ 赋予指向 Ω 外的定向, 则 $\forall P, Q, R \in C^1(\Omega)$, 有

$$\iint_{\partial\Omega} P\,dy\,dz + Q\,dz\,dx + R\,dx\,dy = \iiint_\Omega \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right) dx\,dy\,dz$$

3. Stokes Thm: 设 $M \subseteq \mathbb{R}^n$ 是一个紧致的 k 维流形, 设 w 是光滑的 $(k-1)$ 形式.

则 $\int_{\partial M} w = \int_M dw$

Stokes 公式: 设 $S \subseteq \mathbb{R}^3$ 是定向曲面, 对 ∂S 赋予如下定向, 则 $\forall P, Q, R \in C^1(S)$, 有

$$\int_{\partial S} P\,dx + Q\,dy + R\,dz = \iint_S \det \begin{pmatrix} dy\,dz & dz\,dx & dx\,dy \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{pmatrix}$$

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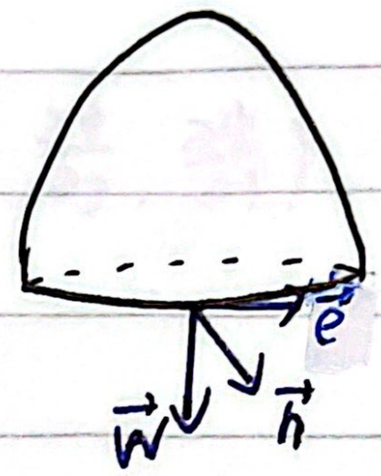
具体写出∂S正定向：

- 右拇：TS中$\perp \partial S$指向S外 $\vec{w}$
- 右食：∂S正定向 $\vec{e}$
- 右中：S定向系统的法向量$\vec{n}$ $\vec{n}$



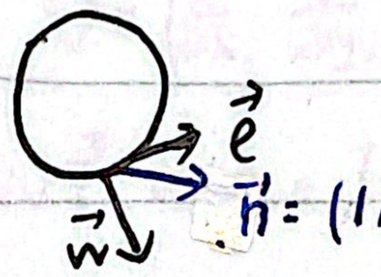
($\vec{e}$的方向是$\vec{n}\times\vec{w}$的方向)

e.g.1 $L=\{x^2+y^2+z^2=1,\ x+y+z=1\}$

定向：$z=+\infty$看为逆，求 $\int_L \underbrace{xy}_{P}dx+\underbrace{yz}_{Q}dy+\underbrace{xz}_{R}dz$

视S为切痕 $T=\{x^2+y^2+z^2\le 1,\ x+y+z=1\}$

为T定向，使∂T正定向$=\vec{L}$



$\vec{n}=(1/\sqrt{3},\ 1/\sqrt{3},\ 1/\sqrt{3})$

$\Rightarrow$原$=\int_{\partial T} xydx+yzdy+xzdz$

$=\iint_T \det\begin{pmatrix} dydz & dzdx & dxdy \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & yz & xz \end{pmatrix} = \iint_T -ydydz-zdzdx-xdxdy$

<u>正型曲面</u>

$\iint_T (-y,-z,-x)\,d\vec{S} = \iint_T (-y,-z,-x)\cdot\vec{n}\,dS = \iint_T \frac{-(x+y+z)}{\sqrt{3}}=1\,dS$

$= -\frac{1}{\sqrt{3}}\,\mathrm{area}(T) = -\frac{1}{\sqrt{3}}\,\pi\left[1^2-(\sqrt{3})^2\right] = \frac{-2\sqrt{3}\pi}{9}$

e.g.2 (Gauss)

$I=\iint_{S\text{定向指向}S\text{外}} \frac{xdydz+ydzdx+zdxdy}{(x^2+y^2+z^2)^{\frac{3}{2}}}$ (设$\Omega=S$围成的3维区域)

① $\vec{0}$在S外

$I \overset{\text{Gauss公式}}{=\!=\!=} \iiint_\Omega \frac{\partial}{\partial x}\left(\frac{x}{(\ )^{\frac{3}{2}}}\right)+\text{etc}=0$

② $\vec{0}$在S内

设 $V=\Omega\setminus\{x^2+y^2+z^2<\varepsilon^2\}$

$\iint_{\partial V} \overset{\text{Gauss}}{=\!=\!=} \iiint_V = 0$

$\vec{n}=(x,y,z)/\varepsilon$

$\iint_{\partial V} = \underbrace{\iint_S}_{=I} + \iint_{x^2+y^2+z^2=\varepsilon^2}$ (向内定向) $\Rightarrow I=\iint_{x^2+y^2+z^2=\varepsilon^2}$ (向外定向) $\frac{xdydz+\text{etc}}{\varepsilon^3} = \iint_{x^2+y^2+z^2=\varepsilon^2}(x,y,z)\cdot\vec{n}\,dS/\varepsilon^3$ (正型曲面)

$=\frac{1}{\varepsilon}\iint dS = 4\pi$

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五. 场论

场 < 标量场 $f: V \to \mathbb{R}$
场 < 矢量场 $\vec{F} = (P, Q, R) : V \to \mathbb{R}^3$

1. 标量场 ϕ

- $\text{grad}\phi \perp$ 等高面 $\quad \text{grad}\phi = \nabla\phi = \left(\frac{\partial\phi}{\partial x}, \frac{\partial\phi}{\partial y}, \frac{\partial\phi}{\partial z}\right)$
- $\int_L \nabla\phi \cdot d\vec{r} = \phi(\text{终点}) - \phi(\text{起点})$

2. 矢量场 $\vec{F} = (P, Q, R)$

2.1 Gauss: $\iint_{\partial\Omega} \vec{F} \cdot d\vec{S} = \iiint_\Omega \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right) dvol$

定义 $\vec{F}$ 的 散度 $\text{div}\vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$ 形式化 $\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)(P, Q, R)$

$= \nabla \cdot \vec{F} \qquad \nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)$

$\Rightarrow$ Gauss 可改述为: $\iint_{\partial\Omega} \vec{F} \cdot d\vec{S} = \iiint_\Omega \text{div}\vec{F}\ dvol = \iiint_\Omega \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right) dvol$

2.2 Stokes: $\int_{\partial S} \vec{F} \cdot d\vec{r} = \iint_S \det \begin{pmatrix} dy\,dz & dz\,dx & dx\,dy \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ P & Q & R \end{pmatrix}$

定义 $\vec{F}$ 的 旋度场 $\text{curl}\vec{F} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}, \text{etc}\right) = \det \begin{pmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ P & Q & R \end{pmatrix}$

$= \nabla \times \vec{F}$

$\Rightarrow$ Stokes 可改述为: $\int_{\partial S} \vec{F} \cdot d\vec{r} = \iint_S (\text{curl}\vec{F})\, d\vec{S} = \iint_S \det \begin{pmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ P & Q & R \end{pmatrix}$

物理意义: 场 $\vec{F}$ 在点 $\vec{x}$ 处绕 $\vec{n}$ 旋转快慢程度

$\triangleq \lim_{\varepsilon \to 0^+} \frac{\int_{\partial D_\varepsilon} \vec{F} \cdot d\vec{r}}{\text{area}(D_\varepsilon)}$ Stokes $\lim_{\varepsilon \to 0^+} \frac{\iint_{D_\varepsilon} (\text{curl}\vec{F}) \cdot d\vec{S}}{\text{area}(D_\varepsilon)} = \lim_{\varepsilon \to 0^+} \frac{\iint_{D_\varepsilon} (\text{curl}\vec{F}) \cdot \vec{n}\, dS}{\text{area}(D_\varepsilon)}$

积分中值 $\text{curl}\vec{F}(\vec{x}) \cdot \vec{n} \qquad (D_\varepsilon =$ 以 $\vec{x}$ 为圆心, ε 为半径, 位于 $\perp \vec{n}$ 的平面的圆盘)

2.3 $d\,dw \equiv 0 \leftrightarrow \begin{cases} \text{curl} \circ \text{grad} \equiv 0 \\ \text{div} \circ \text{curl} \equiv 0 \end{cases}$ $\leftrightarrow$ 命题: ① $\vec{A} = \nabla\phi \Rightarrow \nabla \times \vec{A} = \nabla \times (\nabla\phi) = 0$

② $\vec{B} = \nabla \times \vec{A} \Rightarrow \nabla \cdot \vec{B} = \nabla \cdot (\nabla \times \vec{A}) = 0$

$\Rightarrow$ 逆命题 (Poincare 引理) ① 设 A 是 $\mathbb{R}^3$ 中的光滑场, 则 $\nabla \times \vec{A} \equiv 0 \Rightarrow \vec{A} = \nabla\phi\ (\exists \phi)$

② 设 B 是 $\mathbb{R}^3$ 中的光滑场, 则 $\nabla \cdot \vec{B} \equiv 0 \Rightarrow \vec{B} = \nabla \times \vec{A}\ (\exists \vec{A})$

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证：($\mathbb{R}^3$上 A无旋 $\Rightarrow$ A有势)

势能 $\phi(\vec{p}) \triangleq \phi(\vec{p_0}) + \int_{\vec{L}} \vec{A}\cdot d\vec{r}$

求证 $\nabla\phi = \vec{A}$：$\frac{\partial\phi}{\partial x}(x_0,y_0,z_0) = \lim_{\Delta x\to 0} \frac{\phi(\Delta x+x_0,y_0,z_0)-\phi(x_0,y_0,z_0)}{\Delta x} = \lim_{\Delta x\to 0} \frac{\int_{\vec{L_1}}\vec{A}\cdot d\vec{r} - \int_{\vec{L}}\vec{A}\cdot d\vec{r}}{\Delta x}$

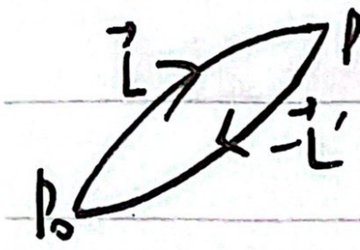
(x_0,y_0,z_0) — $(\Delta x+x_0,y_0,z_0)$

$= \lim_{\Delta x\to 0} \frac{\int_{x_0}^{\Delta x+x_0} A_1(x,y_0,z_0)\,dx}{\Delta x} = A_1(x_0,y_0,z_0)$

其余同理，证毕

(此处计算好好体会！)

补证：验证 ϕ well defined:



$\forall \vec{L'}\neq\vec{L}$, $\int_{\vec{L}}\vec{A}\cdot d\vec{r} - \int_{\vec{L'}}A\cdot d\vec{r} = \int_{\vec{L}\cup(-\vec{L'})}\vec{A}\cdot d\vec{r} = \int_{\partial D}\vec{A}\cdot d\vec{r}$

$\underline{\text{Stokes}}$ $\iint_D (\mathrm{curl}\,\vec{A})\cdot d\vec{S} = 0$

PS: $X\subseteq\mathbb{R}^3$ 不一定对

e.g.1 $C = \{x^2+y^2=1, z=0\}$ $\vec{A} = \left(\frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2}, 0\right)$ (算出 $\nabla\times\vec{A} = \det\begin{pmatrix}\vec{i}&\vec{j}&\vec{k}\\ \partial/\partial x & \partial/\partial y & \partial/\partial z\\ \frac{-x}{x^2+y^2} & \frac{y}{x^2+y^2} & 0\end{pmatrix} = 0$)

但书中使 $A=\nabla\phi$

考虑 $\int_C \vec{A}\cdot d\vec{r}$ $\underline{\text{直接算}}$ $\int_C \frac{-y\,dx+x\,dy}{x^2+y^2}$ $\underline{\text{约化至二维}}$ $\frac{1}{R^2}\underline{\int_{\tilde{C}} -y\,dx+x\,dy} = 2\pi$

$\because A=\nabla\phi$ $= 2\,\mathrm{area}(\tilde{C})$

0 矛盾！

e.g.2 $U=\mathbb{R}^3\setminus 0$ $B = \left(\frac{x}{(x^2+y^2+z^2)^{3/2}}, \frac{y}{\cdots}, \frac{z}{\cdots}\right)$ (算出 $\nabla\cdot\vec{B}=0$)

但书 $\vec{A}$ 使 $B=\nabla\times\vec{A}$ 令 $S=\{x^2+y^2+z^2=1\}$

考虑 $\int_{\partial S}\vec{A}\cdot d\vec{r}$ $\underline{\text{直接用 Stokes}}$ $\iint_S \nabla\times\vec{A}\,d\vec{S} = \iint_S \vec{B}\cdot d\vec{S}$ ~~$\overset{\text{Gauss}}{=}\iiint_{B(1)}$~~ $= 4\pi$

$\because S$无边界 $\therefore = 0$ 矛盾！